

Rishik Gowrishetti

github.com/rishireal | linkedin.com/in/rishikgowrishetti/ | +1 (475) 204-8682 | rg796@cornell.edu | Brookfield, CT

EDUCATION

Cornell University

Ithaca, NY

Bachelor of Arts in Statistical Science, Minor in Computer Science | **GPA: 3.72**

Expected May 2028

Relevant Coursework: Probability and Statistics, Statistics II, Econometrics, Linear Algebra, Multivariable Calculus, Discrete Math, Data Mining and Machine Learning, Object-Oriented Programming and Data Structures, Functional Programming

Involvements: Educational YouTuber (180k+ subscribers); Analyst, Cornell Data Journal; Teaching Consultant, CS 1110; Software, Cornell Entertainment Engineering & Design Club; National Competitor, Cornell Powerlifting Team; Kappa Theta Pi

EXPERIENCE

NASA Goddard Space Flight Center | [Presentation](#)

Greenbelt, MD

Research Assistant, Machine Learning & Uncertainty Quantification

Jun 2026 – Dec 2026

- Proposed and built a Bayesian neural network surrogate (JAX, NumPyro, HMC) that outputs full predictive distributions; beat the group's co-Kriging baseline on a 961k-row ensemble (0.83 vs. 0.47)
- Fixed MCMC non-convergence ($\hat{R}$ -hat 7.8 to 1.01) via input decorrelation, posterior geometry, and noise redesign
- Handled heavy autocorrelation in a 961k-row simulation ensemble; down sampled to 1,500 rows (thinning, stratification, energy-distance twinning) with every feature's mean and SD within 0.5%
- Caught a zero-fill encoding artifact inflating a variable's importance in Sobol/Morris sensitivity analysis; fixed it by splitting the data into storm and quiet regimes
- Researching physics-informed neural networks (PINNs) that embed the governing PDE (Burgers' equation) in the loss

Computational Mechanics & Scientific AI Lab | [GitHub](#) | [Technical Report](#)

Ithaca, NY

Research Assistant

Feb 2026 – May 2026

- Benchmarked an LLM agent as a gradient-free optimizer over a black-box objective (MCP, JAX, Groq) across 300+ trials
- Showed that averaging error over 5 clustered sensors created local minima that trapped the search, dropping success to 26% vs. 75% with full-domain data
- Traced a systematic failure to the agent's bisection heuristic, which cannot reach non-dyadic targets; identified golden-section search and Brent's method as fixes
- Enabled one-step convergence (5/5 trials) by adding analytic physics formulas to the prompt, turning blind bisection into Newton-style updates

Cryptonest (Google Cloud Platform Startup)

Houston, TX

AI/ML & Quantitative Research Intern

Jun 2025 – Aug 2025

- Built a random forest market-regime classifier on daily crypto price time series (rolling returns, volatility, momentum, moving averages) with walk-forward validation; paired it with a max-Sharpe rebalancing strategy that outperformed the firm's static allocation by 1.5× in backtests
- Developed a Portfolio Analyzer agent (MCP, LangGraph) that pulls daily price history over a user-set lookback and automates portfolio diagnostics and allocation recommendations
- Shipped an Efficient Frontier dashboard (Python, Flask, Plotly) with PnL tracking and asset filtering to guide allocation

PROJECTS

TerminalTok: Hybrid Recommender System in OCaml

Oct 2025 – Dec 2025

- Built a hybrid ML recommender combining content-based filtering, collaborative filtering, and matrix factorization
- Implemented matrix factorization from scratch with gradient descent, L2 regularization, and sigmoid-bounded predictions
- Engineered ranking and scoring pipelines using likes, watch time, online training, and top-K neighbor selection

FIRST Robotics Team | Captain, Driver, Programming Lead | [GitHub](#)

Nov 2021 – Jun 2024

- Engineered an autonomous control stack in Java (PID, feedforward, odometry), reducing navigation error by 75%; built a YOLOv5 vision system boosting shot accuracy to 80%

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++, C#, MATLAB, OCaml, Stata, JavaScript, HTML/CSS, Excel

Tools: PyTorch, JAX, NumPyro, scikit-learn, TensorFlow, pandas, NumPy, SciPy, LangGraph, LangChain, OpenAI API, Git